

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q18-1042

Tristate 138 kV  
November 02, 2019

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1042. The Interconnection Customer proposes a 76.2 MW solar photovoltaic facility interconnecting at the Tristate 138 kV substation in the West Virginia area. The request was received on August 22, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 401 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,258,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                  | Contingency                                                 | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------|-------------------------------------------------------------|-------------|--------------|----------|
| Carbon Center 500/345kV Transformer | Blackrock – Shawville Statin 345.0 kV                       | 113.7       | 1,716        | 5.0      |
| W Hill – Fig Garden 138.0 kV        | Siggen – Hyampom 138.0 kV line                              | 112.7       | 180          | 16.2     |
| El Paso – Ravenna 500.0 kV          | Muskingum River – Tidd 500.0 kV                             | 112.6       | 2,948        | 6.0      |
| Tristate – Valemout 345.0 kV        | Sporn – Gavin Power Llc 230.0 kV                            | 111.8       | 1,168        | 19.9     |
| Sporn 345-230 kV Auto               | Sporn Substation – Coulee Ave 345.0 kV line                 | 107.3       | 584          | 5.3      |
| Valemount 138-1 kV TX               | Pence – Valemount 345.0 kV                                  | 105.5       | 448          | 6.5      |
| Burwell 500/345 kV TX               | Allegheny Energy Units 3 4 & 5 345-1 kV Transformer         | 104.9       | 1,191        | 5.5      |
| Valemount – Pence 345.0 kV          | Sporn – Gavin Power Llc 230.0 kV L                          | 104.8       | 1,069        | 19.0     |
| Hilltop 500/345kV Autotransformer   | Hilltop – Talenenergy Martins Crk Llc Williamsport 500.0 kV | 104.1       | 2,218        | 5.7      |
| Contra Costa 3 – W Hill 138.0 kV L  | Siggen – Hyampom Substation 138.0 kV                        | 103.4       | 208          | 16.2     |
| Bland – Tesoro 138.0 kV             | Valemount Substation – Tristate Substation 345.0 kV         | 102.5       | 236          | 10.3     |
| Tesoro – Bowles 138.0 kV            | Valemount – Tristate 345.0 kV                               | 101.7       | 250          | 10.3     |
| Muskingum iv – Tidd 500.0 kV        | El Paso – Ravenna 500.0 kV                                  | 101.7       | 2,844        | 8.3      |
| Bowles – Fig Garden 138.0 kV        | Valemount – Tristate 345.0 kV                               | 101.6       | 253          | 10.3     |
| Tristate – Contra Costa 3 138.0 kV  | Siggen – Hyampom 138.0 kV                                   | 101.5       | 231          | 16.2     |

| Monitored Facility                              | Contingency                   | Loading (%) | Rating (MVA) | DFAX (%) |
|-------------------------------------------------|-------------------------------|-------------|--------------|----------|
| Davis Rd – Hagan 138.0 kV                       | Valemount – Tristate 345.0 kV | 101.2       | 238          | 6.1      |
| Siggen – Hagan 138.0 kV line                    | Tristate – Valemount 345.0 kV | 100.9       | 240          | 6.1      |
| Simpson Stuart Tap – Medora Substation 345.0 kV | Medora – Hampton Rd 345.0 kV  | 100.3       | 1,452        | 5.6      |

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                  | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Carbon Ctr 500/345 kV XFMR                | Replace transformer with higher-rated unit  | \$34,622,000  | 1% (\$323,000)        |
| W Hill – Fig Garden 138.0 kV              | Rebuild line with higher-capacity conductor | \$2,508,000   | 22% (\$551,000)       |
| El Paso – Ravenna 500.0 kV                | Rebuild line with higher-capacity conductor | \$152,151,000 | 1% (\$848,000)        |
| Valemount – Tristate Substation 345.0 kV  | Rebuild line with higher-capacity conductor | \$115,490,000 | 1% (\$1,594,000)      |
| Sporn 345/230 kV XFMR                     | Replace transformer with higher-rated unit  | \$20,600,000  | 7% (\$1,493,000)      |
| Valemount 138/1 kV Auto                   | Replace transformer with higher-rated unit  | \$6,047,000   | 2% (\$121,000)        |
| Burwell 500/345 kV TX                     | Replace transformer with higher-rated unit  | \$25,312,000  | 1% (\$223,000)        |
| Pence – Valemount 345.0 kV                | Reconductor line                            | \$10,479,000  | 1% (\$150,000)        |
| Hilltop 500/345kV Autotransformer         | Replace transformer with higher-rated unit  | \$41,021,000  | 1% (\$321,000)        |
| Contra Costa 3 – W Hill 138.0 kV          | Reconductor line                            | \$1,116,000   | 22% (\$245,000)       |
| Tesoro – Bland 138.0 kV L                 | Reconductor line                            | \$1,508,000   | 14% (\$204,000)       |
| Tesoro – Bowles 138.0 kV L                | Reconductor line                            | \$3,668,000   | 14% (\$496,000)       |
| Muskingum iv – Tidd 500.0 kV              | Reconductor line                            | \$99,840,000  | 1% (\$696,000)        |
| Bowles – Fig Garden 138.0 kV              | Reconductor line                            | \$2,209,000   | 14% (\$299,000)       |
| Contra Costa 3 – Tristate 138.0 kV        | Reconductor line                            | \$1,778,000   | 22% (\$391,000)       |
| Davis Rd – Hagan 138.0 kV                 | Reconductor line                            | \$6,731,000   | 100% (\$6,731,000)    |
| Siggen – Hagan 138.0 kV line              | Reconductor line                            | \$1,430,000   | 100% (\$1,430,000)    |
| Simpson Stuart Tap – Medora 345.0 kV line | Reconductor line                            | \$16,270,000  | 1% (\$142,000)        |
|                                           | Total allocated to project                  |               | \$16,258,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.